Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 05)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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7	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 compares organic and inorganic fertilisers.

0 1 . 1 Complete Table 1 by giving two advantages and two disadvantages of each type of fertiliser.

[6 marks]

Table 1

Fertiliser	Advantages	Disadvantages
Organic (manure, sewage sludge, compost)		
Inorganic (eg ammonium nitrate, NPK)		

0 1 . 2 Explain how hydroponic cultivation maximises productivity and give **two** disadvantages.

[4 marks]

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0 2 . 1 Describe a polyculture system using several species of carp and explain how it increases total productivity.

[4 marks]

0 2 . 2 Explain how integrated multi-trophic aquaculture (IMTA) and aquaponics reduce pollution while increasing food production.

[4 marks]

0 2 . 3 Explain why tuna aquaculture cannot yet replace tuna fishing.

[2 marks]

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0 3 . 1 Describe coppicing and explain how it increases the biodiversity of a woodland.

[4 marks]

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0 3 . 2 Explain how the reintroduction of grey wolves to Yellowstone National Park increased biodiversity.

[4 marks]

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0 3 . 3 Explain why culling or removing species may be necessary in habitat management.

[2 marks]

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0 4 . 1 Explain why the potential to increase food supplies by cultivating new land is limited.

[2 marks]

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0 4 . 2 Explain why importing food may reduce the availability of food in the area where it is produced.

[2 marks]

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0 4 . 3 Explain why organic agriculture is more than simply 'not using pesticides and artificial fertilisers'.

[2 marks]

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0 4 . 4 Evaluate the strategies that could be used to make agriculture more sustainable.

[9 marks]

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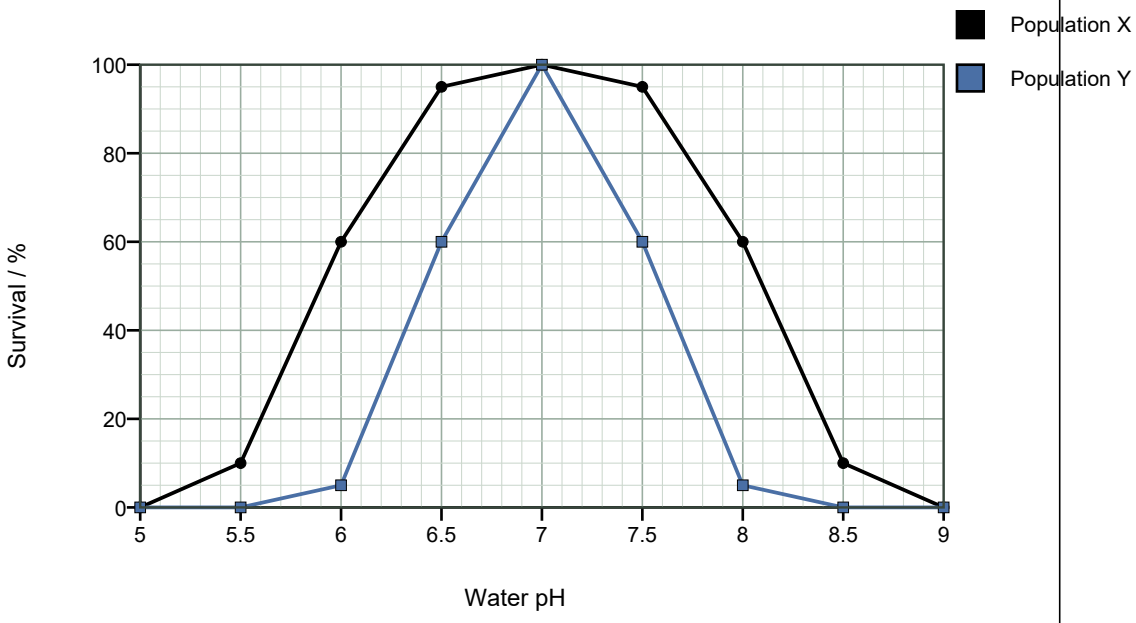
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0 5 Figure 1 shows the percentage survival of two populations of a freshwater snail species (X and Y) at different water pH values.

Figure 1



0 5 . 1 Explain what is meant by the term **range of tolerance** and use Figure 1 to compare the ranges of tolerance of populations X and Y.

[3 marks]

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0 5 . 2 A stream is acidified by mine drainage from pH 7.0 to 6.0. Use Figure 1 to explain which population is more likely to survive and why.

[3 marks]

0 5 . 3 Explain how an understanding of a species' range of tolerance helps in conservation management.

[2 marks]

0 5 . 4 Give **two** biotic factors that affect the distribution of a species.

[2 marks]

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0 6 . 1 Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

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0 6 . 2 Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

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0 6 . 3 Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

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0 7 . 1 Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

[1 mark]

A Environmental DNA (eDNA) analysis ☐

B Radio tracking ☐

C Lincoln index (mark-release-recapture) ☐

D Leg rings ☐

0 7 . 2 Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

[2 marks]

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0 7 . 3 Explain how DNA databases could help to control the illegal trade in elephant ivory.

[2 marks]

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0 8 Table 1 shows the number of bird species and butterfly species recorded in three woodlands in the same region.

Table 1

Woodland	Management	Bird species	Butterfly species
A	Ancient oak wood, coppiced on rotation with clearings	42	23
B	Ancient oak wood, unmanaged for 60 years, dense canopy	28	9
C	40-year-old conifer plantation	14	3

0 8 . 1 Use Table 1 and your knowledge to explain the differences in biodiversity between the three woodlands.

[5 marks]

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0 8 . 2 Describe **three** conservation efforts used to protect UK woodlands.

[3 marks]

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0 8 . 3 Explain why a high biodiversity increases the ecological stability of a woodland.

[2 marks]

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0 9 . 1 Explain why introduced species often have a greater effect on island communities than on mainland communities.

[2 marks]

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0 9 . 2 For each of the following, describe an example of an introduced species and explain its effect on an indigenous species.

[6 marks]

Introduced competitor .....

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Introduced predator .....

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Introduced pathogen .....

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0 9 . 3 Explain how hybridisation with an introduced species threatens the red deer and the Scottish wildcat.

[2 marks]

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1 0 . 1 Evaluate the extent to which the principles of the circular economy can be applied to **three** of the following: water supplies, mineral supplies, waste management, pollution control, energy supplies, food production.

[9 marks]

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1 0 . 2 State what is meant by **industrial symbiosis**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.

[25 marks]

OR

1 1 . 2 Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS